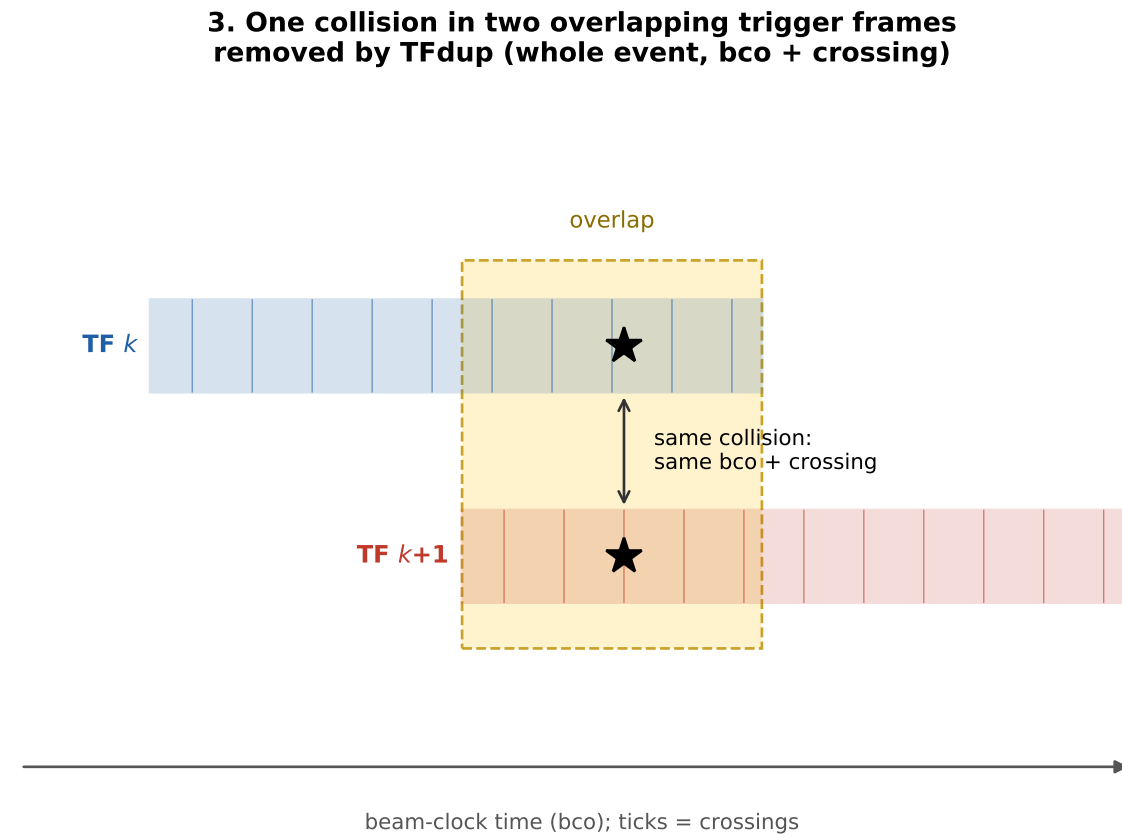
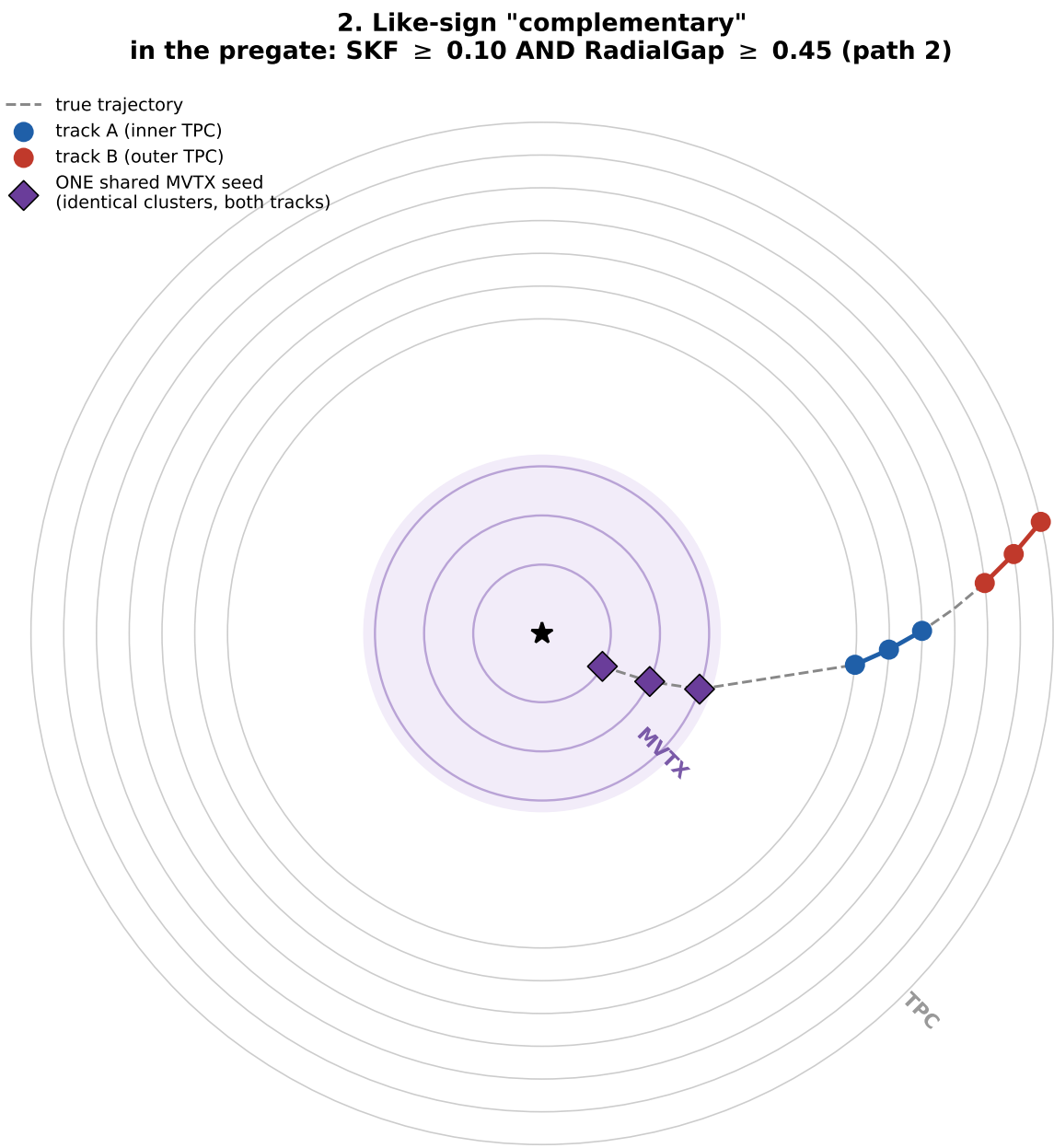
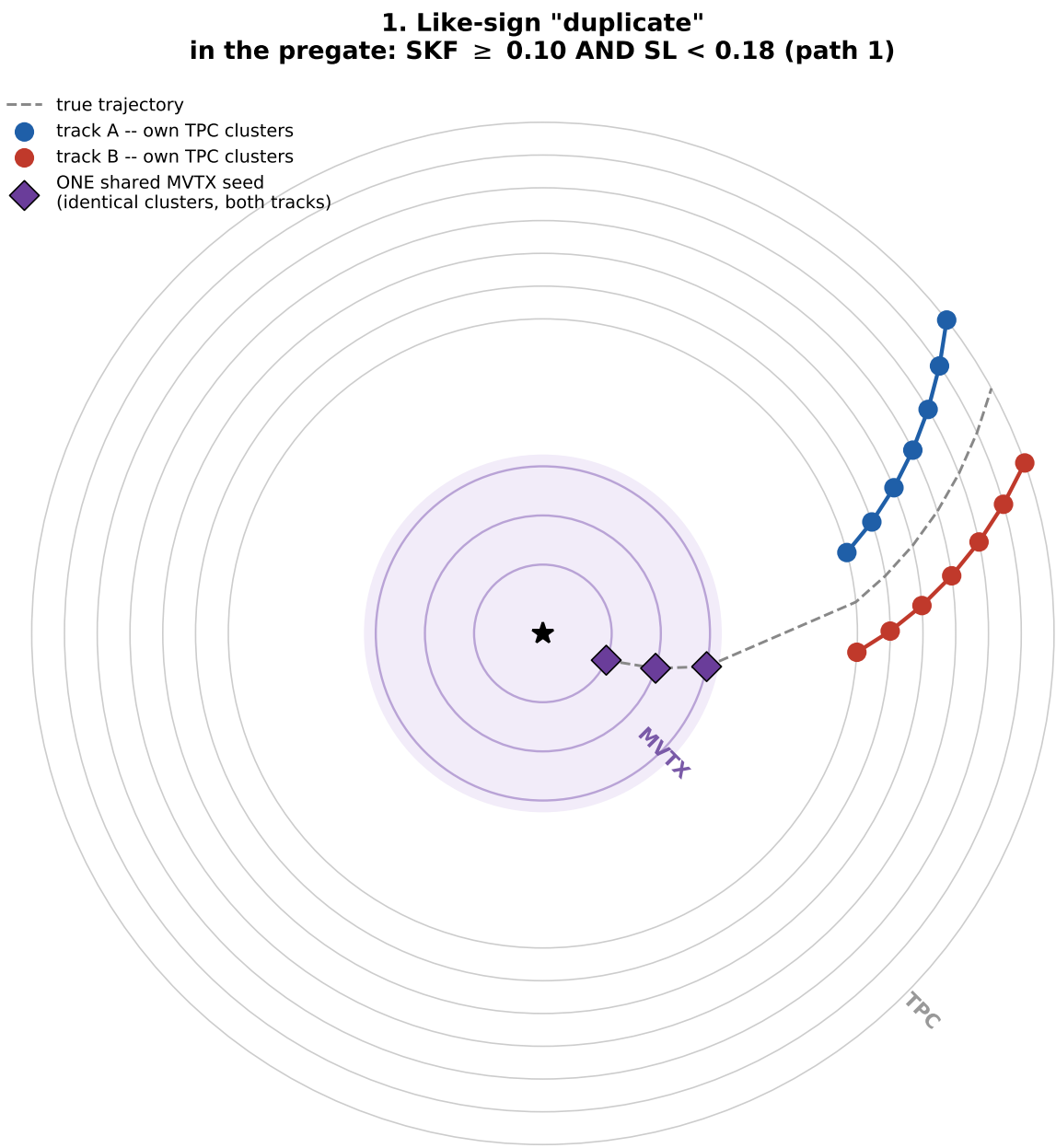




The copy in TF $k+1$ is the SAME event (same tracks, same vertex): kept, it would sit in the mixing pool next to its original and mix with its own copy -- a denominator excess at (0,0) (sec 22.1).

Removes 12.9% of the ana573 events (10M of 79514), most of them one TF later (93%).
First copy read is kept; ana532 has no bco branch (no-op there).

Same charge, same silicon seed, two TPC cluster sets on the SAME layers (but different clusters, ~ 72 mm apart, sec 17.19).

$SL \approx -0.45$ (shares $\approx$ every TPC layer), $RadialGap \approx 0.02$

Pair lands at $(\Delta\eta, \Delta\phi) \approx (0, \Delta\phi_0)$: the fragments share the seed but can differ in p_T , so they are $\Delta\phi_0 = |\text{asin}(aR/p_{T1}) - \text{asin}(aR/p_{T2})|$ apart ($R \approx 6$ cm); the pregate is centred there, its half-width $\max(2^\circ, 3\sigma)$, $\sigma = 0.22^\circ + 0.13 \Delta\phi_0$ (sec 12, 33).

If SL and RadialGap both miss: ≥ 2 MVTX layers with the same cluster still flag it (path 3, sec 8/15).

Removes 1.60% of the accepted tracks (path 1) + 0.14% (path 3).

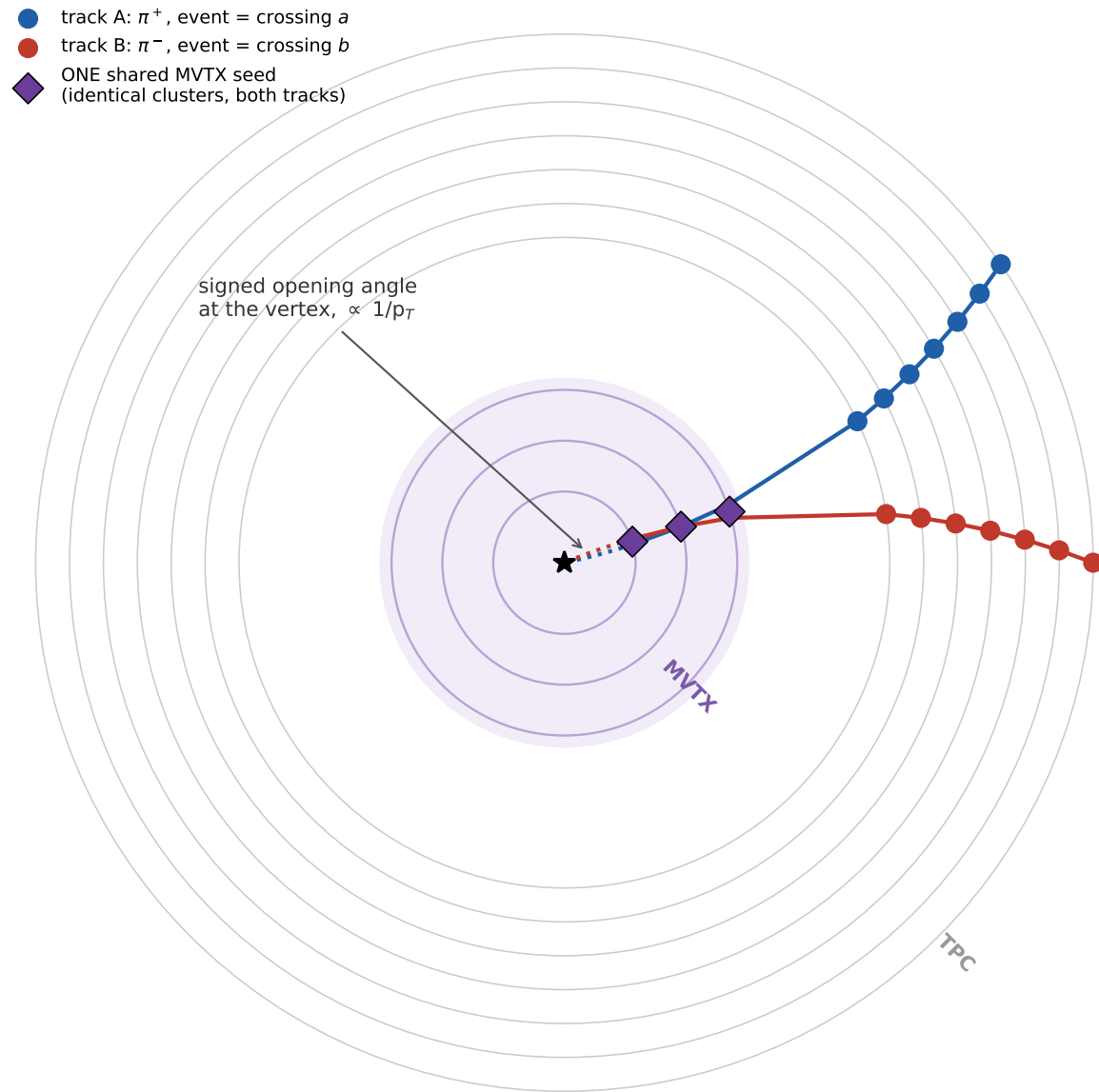
Same charge, same silicon seed, one trajectory's TPC clusters torn into DISJOINT inner/outer segments (STAR fig-1b picture).

$SL = +1$ (no shared pad-rows: SL is blind), $RadialGap \approx 0.53$

Survived the SL cut as a two-lobe residual ridge at (0,0).

Rare in ana573: 0.006% of the accepted tracks.

4. One silicon seed, two opposite-charge TPC tracks ULS veto (same crossing) / XTF (other crossing of the TF)

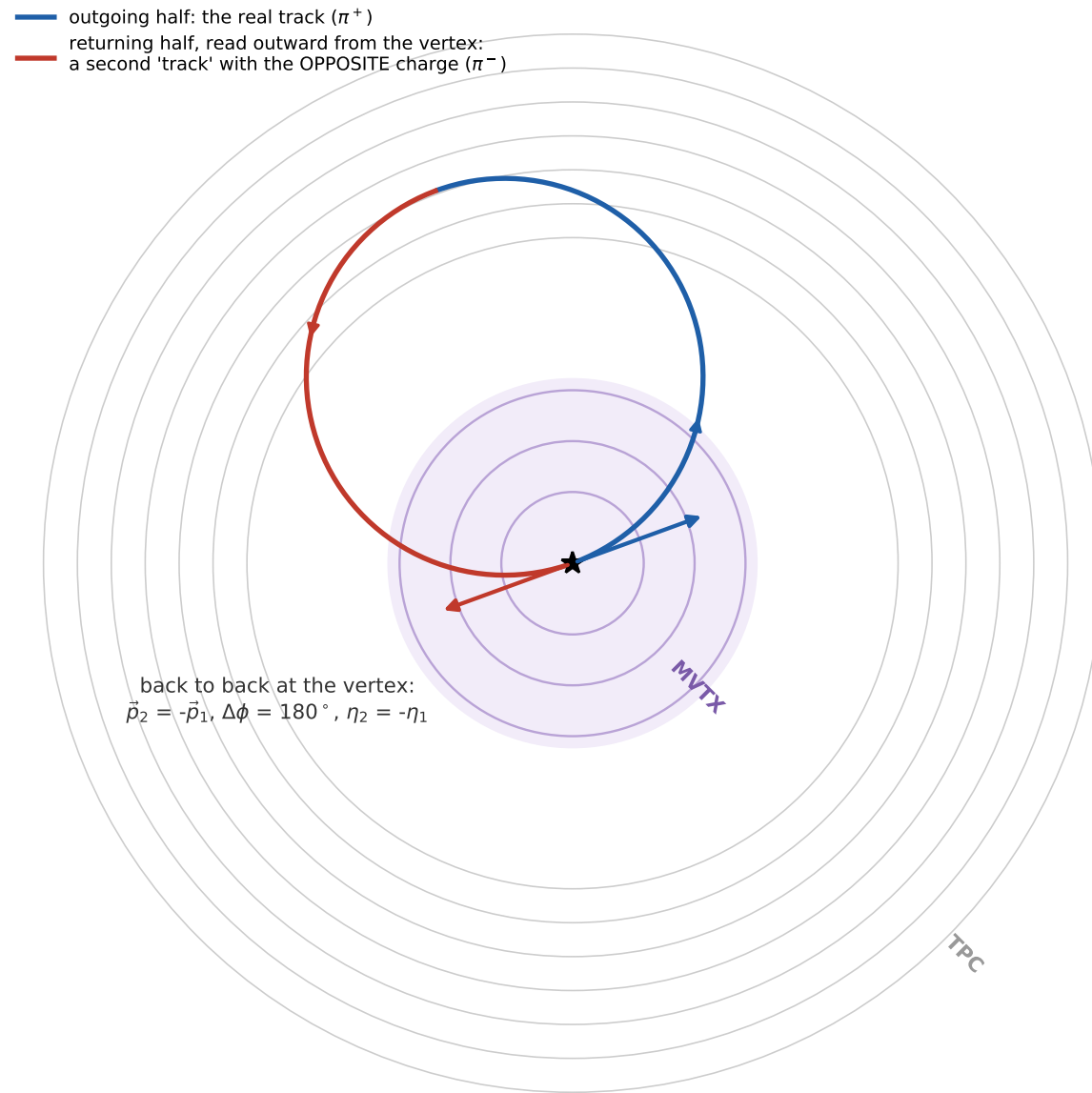


The matcher attaches one silicon seed to several TPC seeds; each combination can get its own crossing, so the copies sit in the same event OR in DIFFERENT events of one trigger frame (TF).

$$\pi^+ \pi^-: \Delta\phi \approx 1.25^\circ + 1.13^\circ \text{ GeV}/p_T \text{ (the ridge)}$$

Same crossing: ULS veto (SiSplitScore ≥ 0.05 , any angle): 1.10% of the tracks.
Other crossing: XTF keeps the copy whose crossing = its INTT time bucket: 1.96% of the tracks. Adjacent TFs: kept out of the mixing.

5. A looper: one curling track seen twice removed by the looper veto (the worse half)

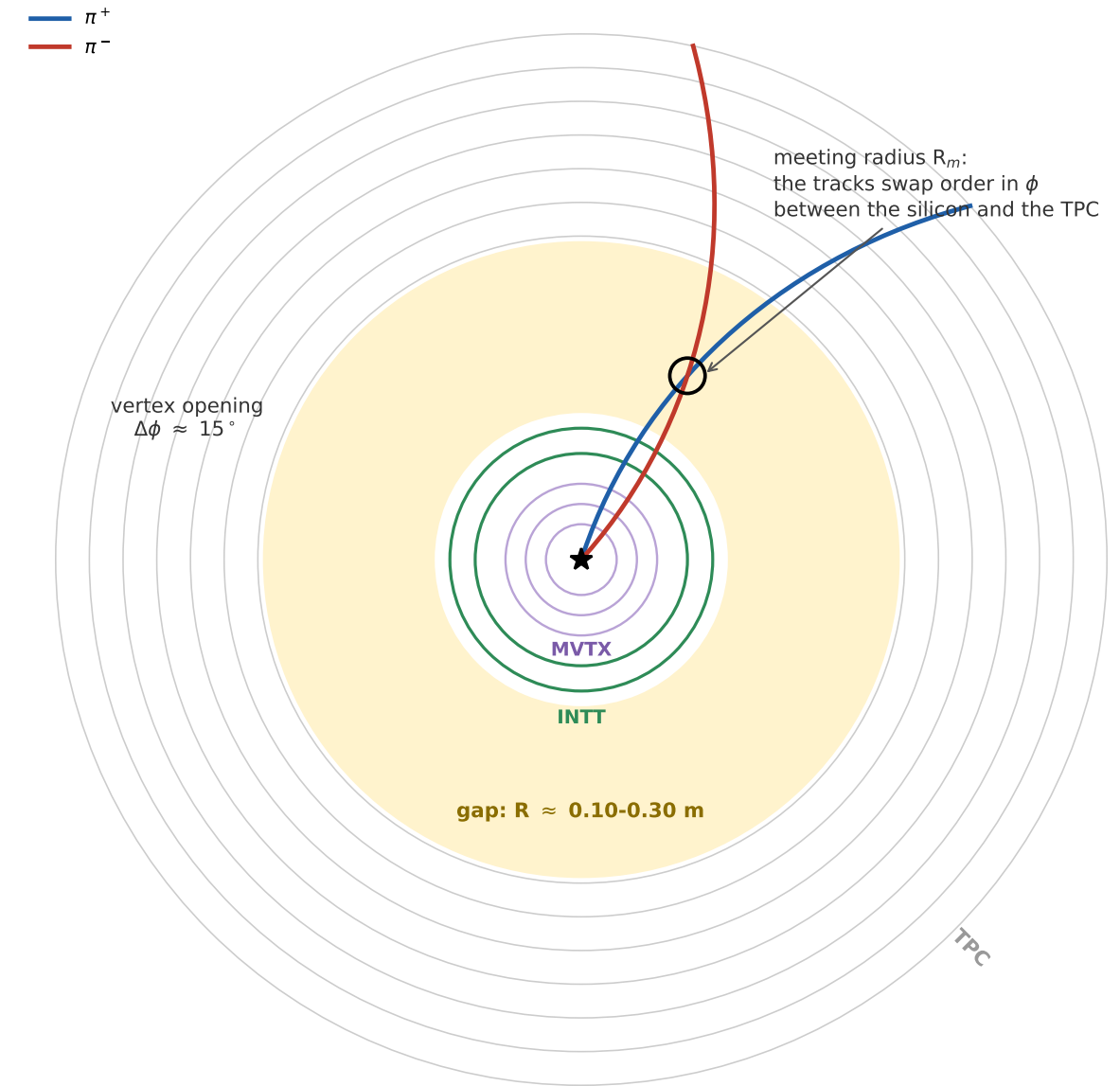


$p_T < \sim 0.17 \text{ GeV}/c$: the circle ($2r < 0.78 \text{ m}$) closes inside the TPC and the track comes back through the centre. Read outward, the returning half is an opposite-charge 'track', exactly back to back, with the same $|p|$.

Signature (50M, sec 33.4): $|\vec{p}_1 + \vec{p}_2| \approx 0$, $M_{inv}(\pi\pi) = 0.35\text{-}0.45 \text{ GeV}$; the lower edge is the $0.1 \text{ GeV}/c$ p_T floor (ana532 starts at 0.2: no loopers).

Veto: OS, both $|p| < 0.20 \text{ GeV}/c$, $\Delta\phi \geq 170^\circ$, $|\vec{p}_1 + \vec{p}_2|/(|p_1| + |p_2|) < 0.08$.
Removes 0.010% of the tracks (0.08% at p_T 0.1-0.2); the $\pi^+ \pi^-$ away-side spike at $dy \approx 0$ goes $+0.021 \rightarrow +0.005$ ($= \pi^+ \pi^+$). Not cosmics.

6. The $\pm 15^\circ$ ULS notch: two tracks that meet in the INTT-TPC gap LOST in the silicon-TPC matching -- watched, not removed



Opposite charges bend TOWARD each other. For $\langle p_T \rangle \approx 0.2\text{-}0.8 \text{ GeV}/c$ a 15° vertex opening puts the meeting point at $R_m \approx 0.10\text{-}0.30 \text{ m}$, where the TPC seeds are matched to silicon seeds: the pair is lost (sec 34.2).

ULS sibling pairs with $R_m = 0.15\text{-}0.20 \text{ m}$: -7.6% (10-20° band); LS pairs that meet there (unequal p_T) lose $\sim 3.5\%$ too. Depth $\sim 4\%$ at 10-20°, independent of N_{ch} , ϕ and y ; no Corral cleaner makes it (sec 32-33).

Watched on the notch MONITOR page of every production; for tracking.
A pair-acceptance correction for $R_m = 0.1\text{-}0.3 \text{ m}$ is possible later.

7. A track that crosses the central membrane inside the TPC its TPC clusters are split between south and north

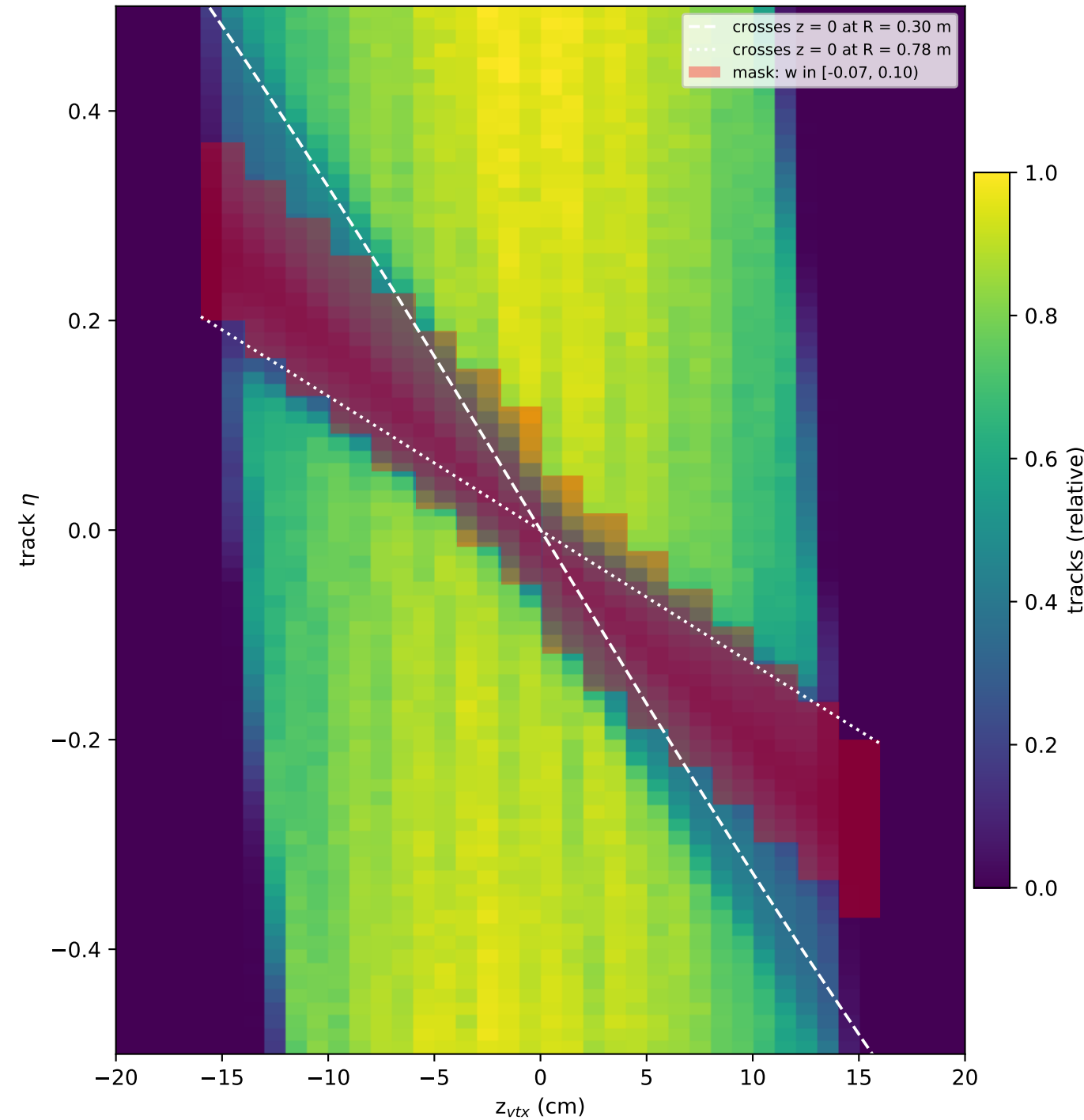
- $\eta = 0.03$: never reaches $z = 0$ in the TPC (fine)
- $\eta = 0.13$: crosses $z = 0$ at $R \approx 0.54$ m -- IN THE HOLE
- $\eta = 0.40$: crosses before the TPC (fine)



Side view (z stretched x5). From a vertex at z_{vtx} , a track crosses $z = 0$ at $R = -z_{vtx}/\sinh \eta$. For R between the TPC inner (0.30 m) and outer (0.78 m) edge its clusters sit on both sides of the membrane: $\eta \approx \text{asinh}(-z_{vtx}/R)$, a band that moves with z_{vtx} ($\approx -0.018 z_{vtx}/\text{cm}$).

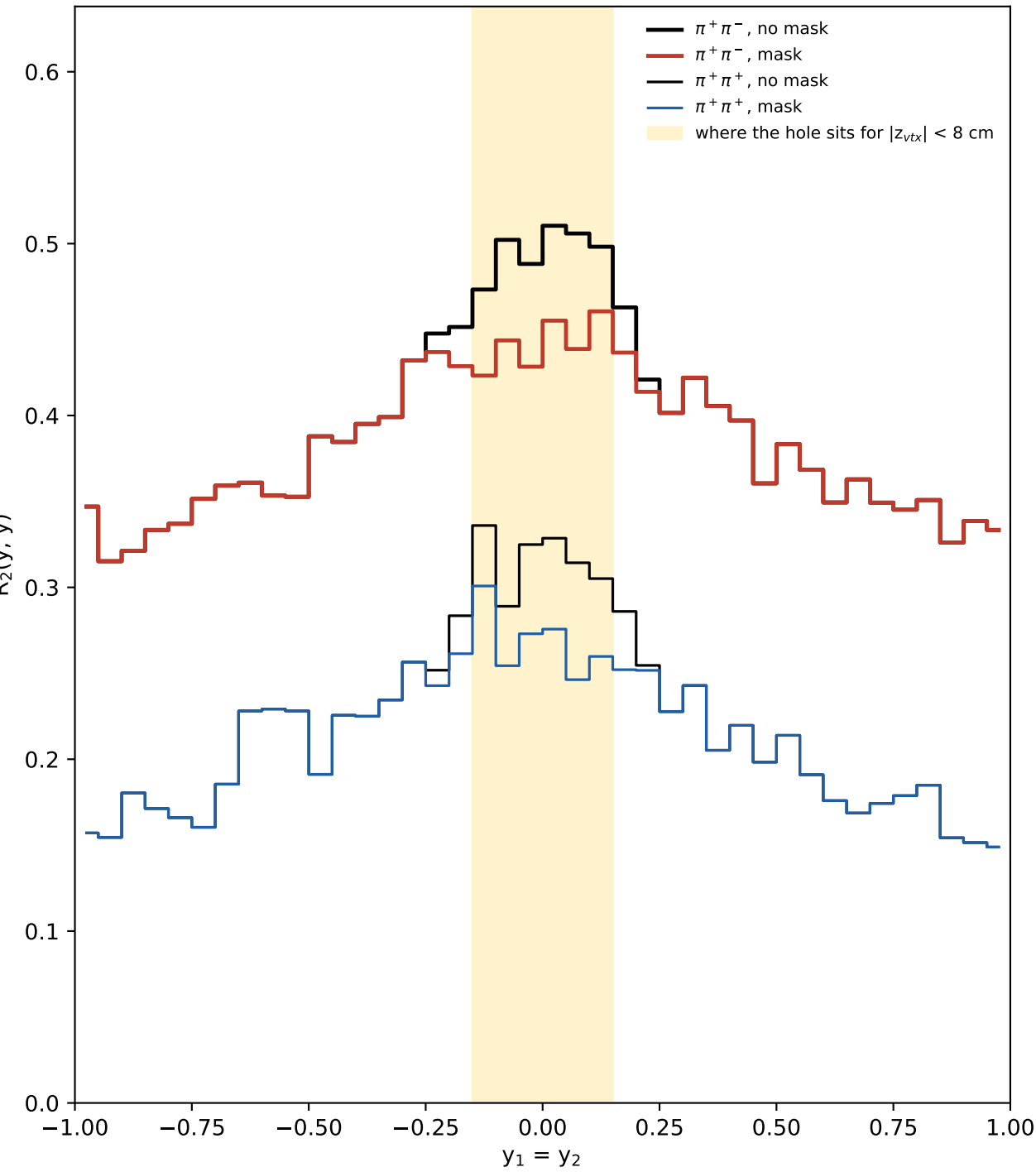
Such tracks are often lost (the hole in η at every z_{vtx} ; ana573 ~3x deeper than ana532). In SOME events they are not: tracks in the hole come in pairs 1.7x more often than any others (F_2 2.20 vs 1.30), not tied to the crossing, the multiplicity or close pairs.

8. The hole in the data (10M events, no mask) and the mask that follows it



$w = (\eta + 0.018 z_c) \text{sign}(-z_c)$, z_c = the zvtx-slice centre: the hole at $w \approx 0$.
The mask drops tracks with w in $[-0.07, 0.10]$: 2.79% of the tracks, from S and M alike.
Fixed within each 2-cm slice, so every event of a slice has the same acceptance.

9. $R_2(y_1, y_2)$ along $y_1 = y_2$: the peak at 0 is the hole removed by the CM mask (default; nocmmask = off)



Bump = $R_2(y, y)$ at $|y| < 0.2$ minus at 0.25-0.45 ($\Sigma S/\Sigma M$ over the slices):
 $\pi^+ \pi^-$ 0.082 $\rightarrow$ 0.035, $\pi^+ \pi^+$ 0.079 $\rightarrow$ 0.036 (ana532, no mask: 0.056 / 0.061).
Finer z_{vtx} slices do NOT help (1-cm slices: unchanged). A mask that follows each event's z_{vtx} does not work either: its edge moves inside a slice (first scan).